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TRANSPORT PHENOMENA II, CHE 312

QUIZ 1

CLOSED BOOK, TIME: 15 MINUTES

A liquid with a plug flow velocity profile $v_z(r) = v_0$ is being heated within a cylindrical tube of radius R with a constant heat flux $q_r(z, r = R) = q_R < 0$.

Model assumption:

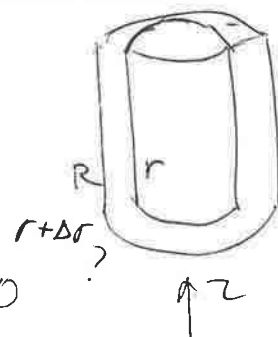
- Steady-State
- Constant physical properties ρ, c_p, k .
- $c_v \approx c_p$ is constant and $\frac{du}{dT} = c_p$
- Heat flux in the axial direction is negligible.
- Negligible viscous heating.
- Fluid enters the heating region of the tube with a uniform temperature $T(z = 0, r) = T_1$

Derive an expression for the bulk (cup-mixing average) temperature of the fluid as a function of z . T_b ?

$$0 = \int_S dS \langle q|n \rangle + \text{convection}$$

$$\underbrace{[\text{heat in}] - [\text{heat out}]}_{\text{conduction}} + \text{convection} = 0$$

$$\underbrace{(r^2 q_r(r)) - ((r+\Delta r)^2 q_r(r+\Delta r))}_{\text{conduction}} + \text{convection} = 0$$



$$\lim_{\Delta r \rightarrow 0} \frac{r^2 q_r(r) - (r+\Delta r)^2 q_r(r+\Delta r)}{\Delta r} + \frac{du}{dT} (V_z(r)) = 0$$

$$\int \left(\frac{d}{dr} (q_r r^2) \right) + \frac{du}{dT} v_z = 0$$

$$q_r \frac{r^3}{2}$$

ok-ish
but wrong

missing details

r - contribution wrong

z - " wrong

math wrong

physical understanding lacking